Assignment Writeup

Main Question

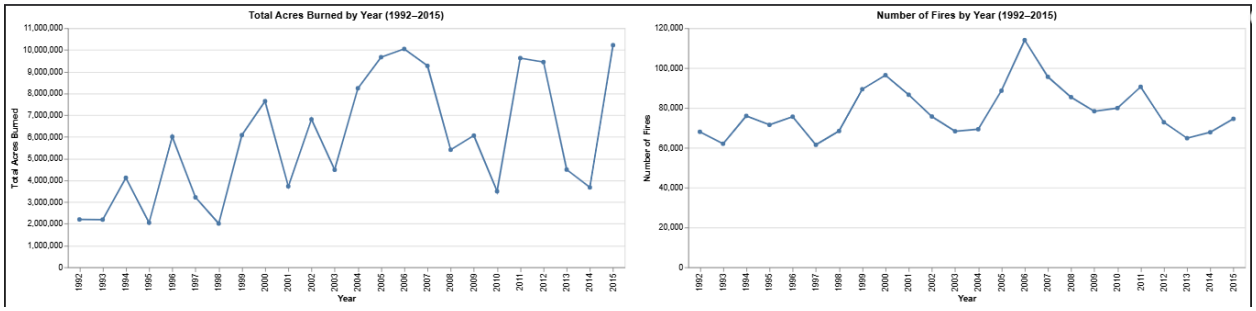
Have U.S. wildfires become a bigger problem in terms of size and in terms of frequency? Size will be measured in the number of acres burned and also by the number of reported wildfires across the United States from 1992 to 2015 by year.

The Data

Data was obtained from Kaggle with the citation listed below. The dataset itself is from an SQLite database that contains 39 columns and 1,880,464 rows of recorded wildfire reports in the United States over a 24 year span (1992-2015). The dataset includes variables such as date or reported fire, the cause of the fire, the approximate coordinates as well as the corresponding county and state, the responding agency, the amount of acres burned, the category of the fire (measuring severeness by total acres burned by that specific fire) and other detailed variables. Some limitations to this dataset is that it doesn't explicitly state any destruction caused by the wildfires and that the most recent years for this dataset are 10 years old. However, my father is in the fire service and I am a huge outdoor lover so this dataset really sparked my interest and it was a project I can put on my resume that is different from other projects and emphasizes time series data. The original records come from all agencies, from local to federal and was a part of the third update of a publication to support the national Fire Program Analysis (FPA) system with the data (when applicable) being transformed to fit the standards of the National Wildfire Coordinating Group (NWCG). The publication mentioned earlier was written by Karen Short and is titled "Spatial wildfire occurrence data for the United States 1992-2015". Below are citations for both sources:

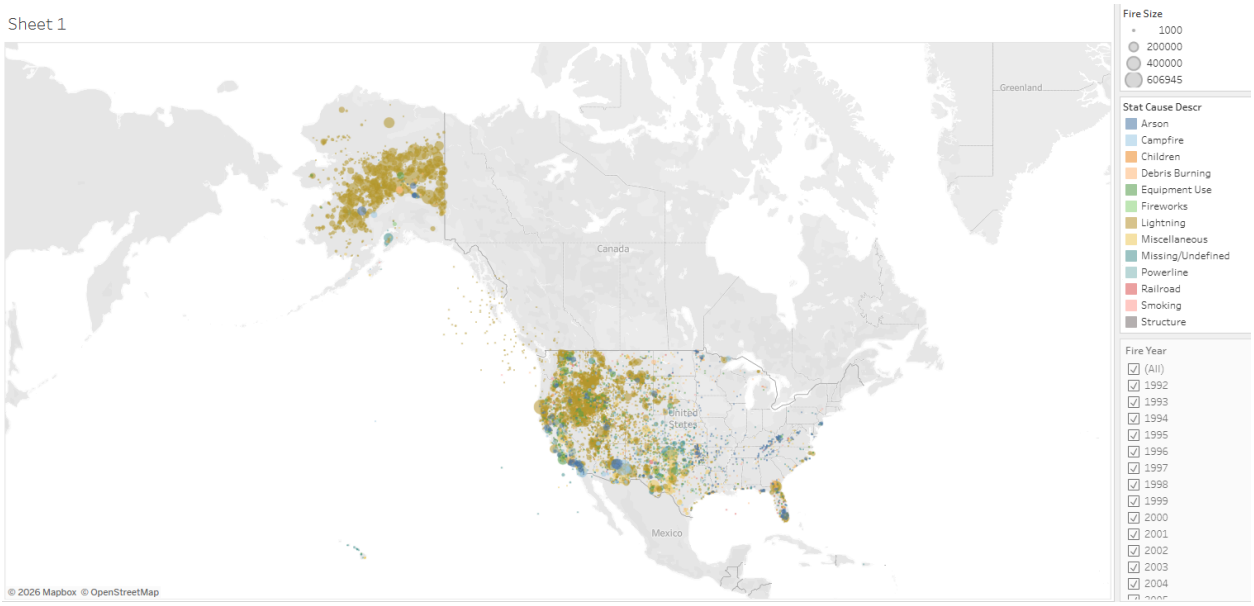
Tatman, Rachael. *1.88 Million US Wildfires*. Kaggle. https://www.kaggle.com/datasets/rtatman/188-million-us-wildfires (accessed October 28, 2026).
Short, Karen C. 2017. *Spatial wildfire occurrence data for the United States, 1992-2015 [FPA_FOD_20170508]*. 4th Edition. Fort Collins, CO: Forest Service Research Data Archive. https://doi.org/10.2737/RDS-2013-0009.4

Chart 1

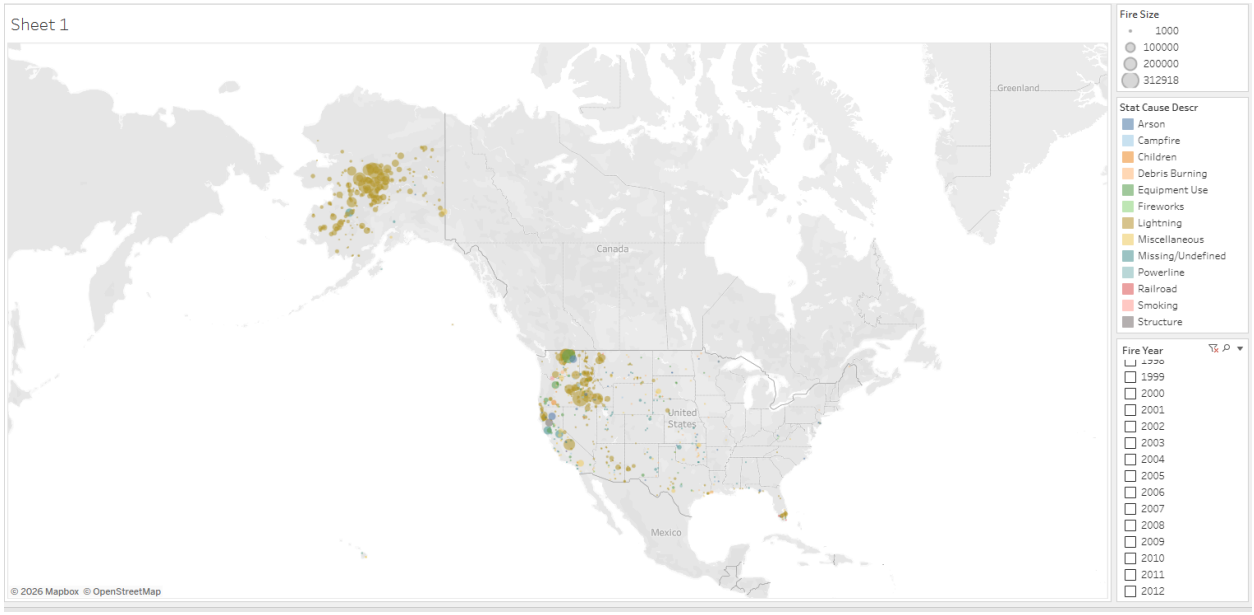


The first chart shows the number of acres burned and the number of reported fires for each year from 1992 to 2015. From 1992 to 2003, acres burned rise and fall in a fairly consistent pattern of spikes. From 2003 to 2006, acres burned only increased, and from 2007 to 2010 they mostly declined. But, the year 2010 does not fall back to the level of the early years as it is the lowest point from 2006 to 2015, yet it is still almost twice the acreage of some earlier years. Then, in 2011, 2012 and 2015, acres burned reached nearly 10,000,000 acres. The number of reported fires tells a different story. It spiked between 1998 and 2000, between 2004 and 2006, and a very mild spike again in 2011, which does match some of the spikes in acres burned. But it did not spike to the extent that I was expecting in 2011, and there is little to no definitive increase in reports in 2012 or 2015. In addition, 1992 and 2015 have roughly the same number of reported fires while 2015 had nearly 5x the amount of acres burned as 1992. What needs to be looked at now is how these fires start and where to see if that can help explain these trends. This visualization was made using Altair

Chart 2



Full Year Dataset

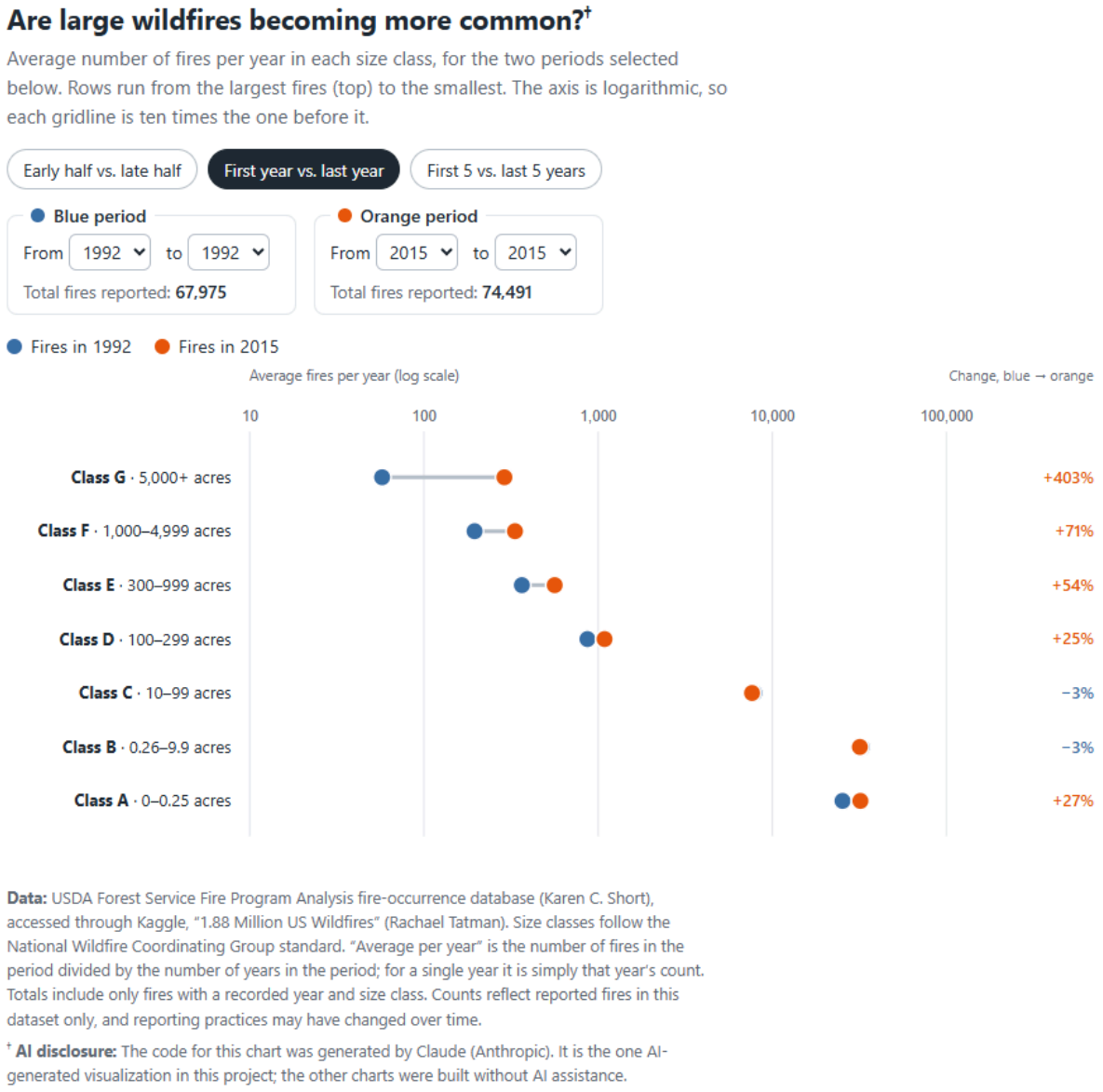


1992 Compared to 2015

Here we have a map of the United States where we can filter by year. Our colored circles recommend the cause of the fire and the relative size of the circle shows the acres burned by that specific fire. An important notice is that we set the size scale to 1000 as keeping any fires under 1000 acres crowded the map too much and made it hard to interpret. We can also hover over any dot and a tooltip will appear with the specific fire size, starting reason and coordinates. Here we can see that lightning is a primary cause of a lot of these larger fires, along with equipment use, arson, campfires and fireworks also making a significant chunk of larger fires started. When we filter by year we can see in 1992 there is a very small amount of large fires in Alaska and the size of these fires are also rather smaller. But when we look at 2015 we see not only a significant

increase in the number of fires but also the relative size of these fires, but there is a slight decrease in the amount of fires seen on the mainland of the United States which is consistent with the first chart showing the burnt acres being greater but the reported sightings being similar. But all of this is just speculation and eyeballing the data, to really see an increase between years or any real change we would need to break it down using the data.

Chart 3



Data: USDA Forest Service Fire Program Analysis fire-occurrence database (Karen C. Short), accessed through Kaggle, "1.88 Million US Wildfires" (Rachael Tatman). Size classes follow the National Wildfire Coordinating Group standard. "Average per year" is the number of fires in the period divided by the number of years in the period; for a single year it is simply that year's count. Totals include only fires with a recorded year and size class. Counts reflect reported fires in this dataset only, and reporting practices may have changed over time.

† **AI disclosure:** The code for this chart was generated by Claude (Anthropic). It is the one AI-generated visualization in this project; the other charts were built without AI assistance.

The final chart compares the number of fires reported in each size class between two periods, which the viewer chooses with the filters above the chart. Each row has a size class,

from Class G (5,000+ acres) at the top to Class A at the bottom, following the NWCG standard. The blue dot is the average number of fires per year in the first period and the orange dot is the second period. In this example, blue is 1992 and orange is 2015 but the user can select a range of dates such as 1992-2000 and 2005-2015. The horizontal axis uses a log scale, because small fires far outnumber large ones and the log scale allows us to fit the chart better for a more friendly user experience, and the percentage on the right shows how much the count changed from blue to orange. Class G fires rose from 58 in 1992 to 292 in 2015, a 403% increase. Also looking at Classes F, E and D had 71%, 54% and 25% more fires in 2015 compared to 1993. The total number of reported fires was also higher in 2015, by roughly 6,500 fires, but that alone doesn't explain the jump in acres burned. Looking at 1993, about 6,000 fewer fires were reported than in 1992, yet acres burned in those two years were similar. Looking at the difference, we see that 1992 had more reported fires in the smaller Classes while 1993 had 12% more fires in Class G. Finally, looking at the first five years of the dataset vs the last five years of the dataset there is only a difference of about 17,000 fires reported (370,345 to 352,965 fires) yet the later five years have 94%, 29%, 10% and 11% increase in the four largest class of fires. This suggests that wildfires have become a bigger threat through larger fires rather than simply more fires.

Conclusion

Looking at everything, the three charts suggest that U.S. wildfires have become a bigger threat mainly because individual fires have grown larger, and not because more fires are being reported. The number of reported fires fluctuates from year to year, with spikes around 1998-2000 and 2004-2006, but 1992 and 2015 are roughly comparable with most years falling between roughly 60,000 and 85,000 reports. Acres burned follow a different pattern. After 2003, the peaks are much higher and the low years stay above the early lows in the 90's and early 2000's. The map shows more large fires in 2015 than in 1992, and the size class comparison shows the biggest increases in the largest classes, with Class G fires rising 403% between 1992 and 2015. Because acres burned depend on both how many fires there are and how big they are, these results point to fire size as the bigger driver of the increase. This analysis has limits. The data ends in 2015, reporting practices may have changed over time, and the dataset doesn't measure damage to homes or lives, so "threat" here means size and frequency only.